

Optiver 



Introduction to Financial Markets, Trading and Execution



Optiver Trading Academy 2026



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Trader

Lecturer

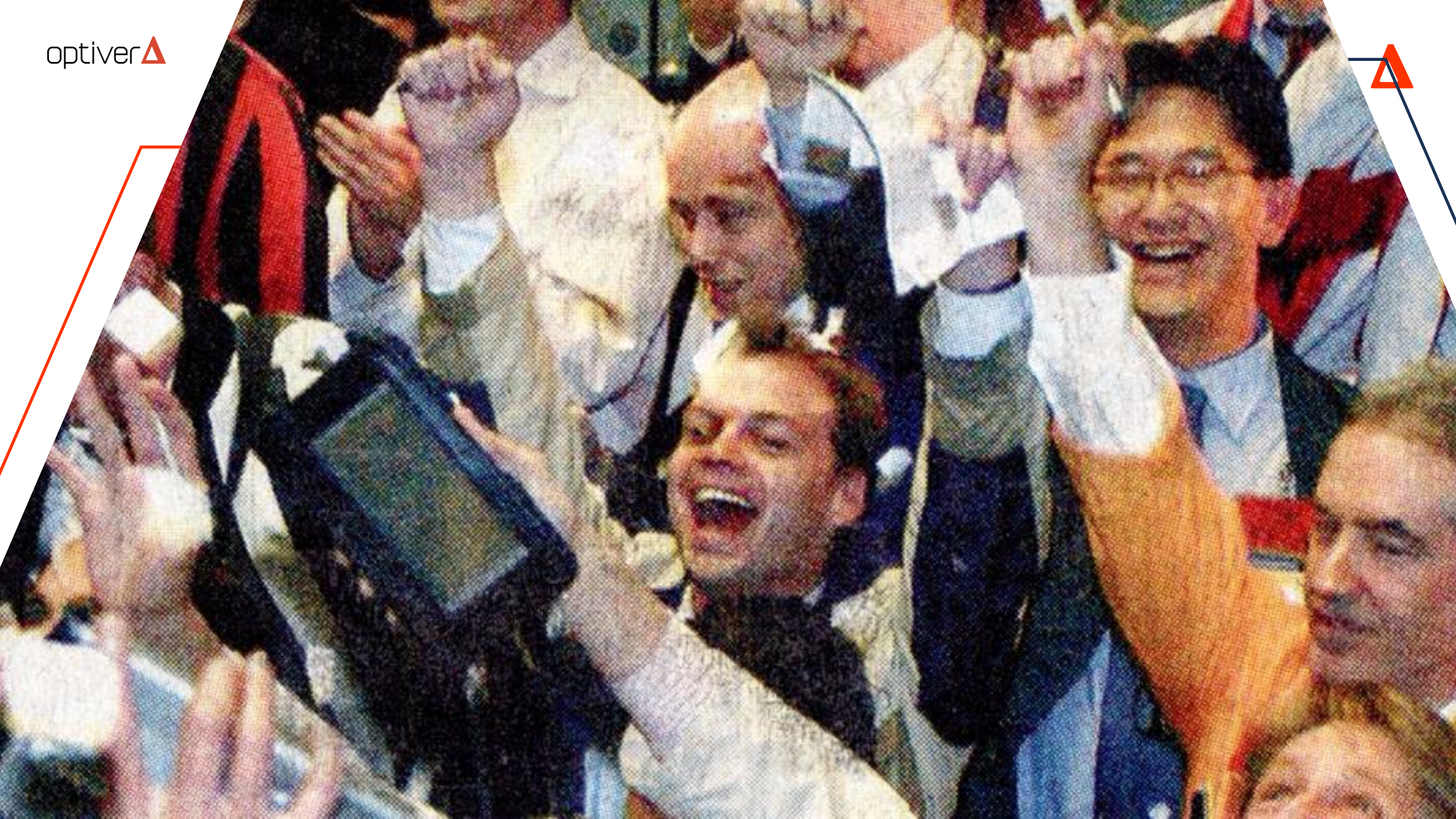


Robbert Pullen

Trader

Trading trainer

Head of academic partnerships





We improve the market



Founded in 1986



By a single trader on
the Amsterdam'
Options Exchange

**Market maker/
liquidity provider**



We stabilize and
improve financial
markets

Privately owned



We are privately
owned by current and
former employees

~ 2400 FTE



We have a global
footprint with 57
nationalities represented

**11 offices across
4 continents**



We are a multi-
national company
with a global footprint

Own capital



We use our own
capital to trade,
at our own risk

We are a global firm





Agenda

01 Financial markets structure

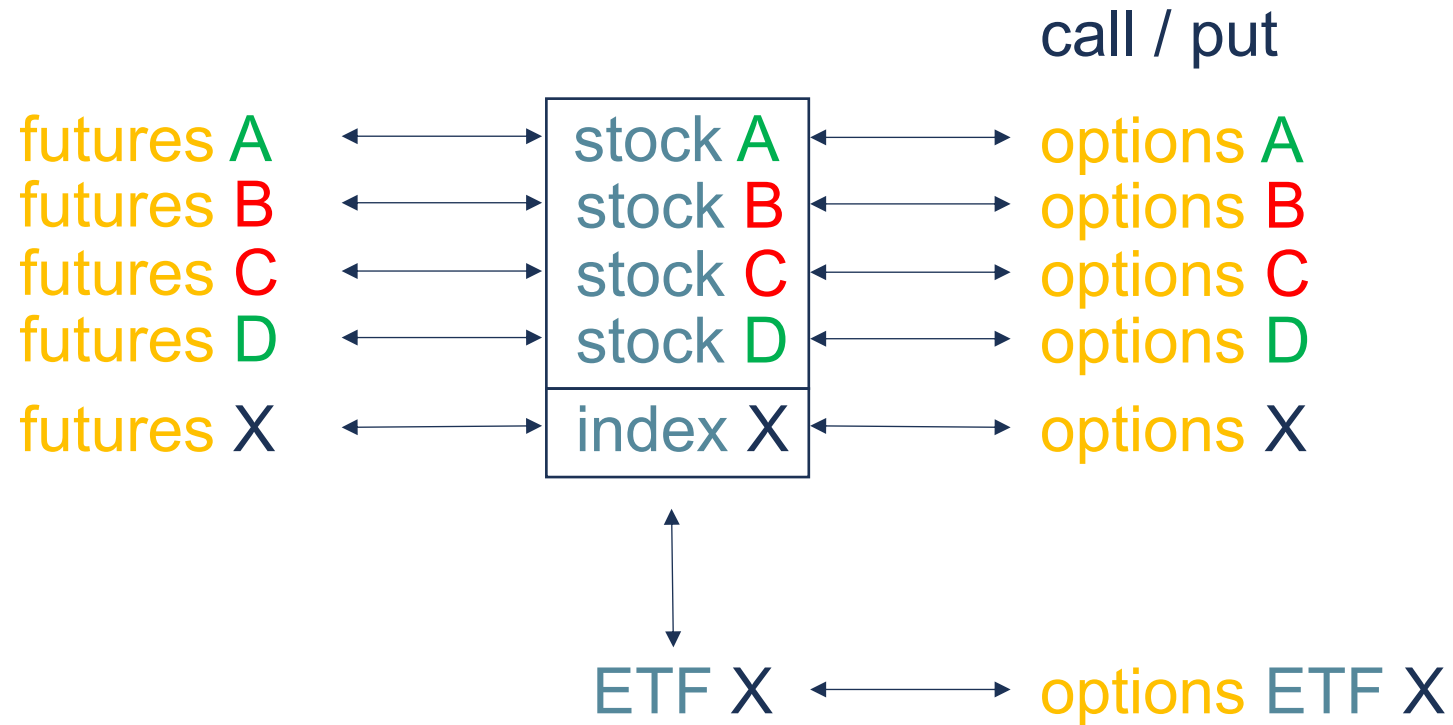
02 Trading and execution



Financial Markets Structure

- Financial instruments
- Market participants
- Liquidity
- Market Making
- Hedge

Financial Instruments



Market Participants



Asset managers
Investment banks
Pension funds
Mutual funds
Retail investors
Others



Directional

Market Makers
Brokers
Intermediaries
Internalisers
High-Frequency Traders
Others



Non-directional /
Market neutral

Liquidity



liquid

bid volume	price	ask volume
	44.67	8539
	44.66	10551
	44.65	3677
	44.64	6800
	44.63	9061
	44.62	6693
	44.61	6573
2049	44.60	
5176	44.59	
1080	44.58	
5000	44.57	
16776	44.56	
2000	44.55	
15349	44.54	
7800	44.53	
11327	44.52	

▲ Volume

▲ Bid-ask spread

illiquid

bid volume	price	ask volume
	44.67	745
	44.66	496
	44.65	
	44.64	400
	44.63	14
	44.62	500
	44.61	
	44.60	
	44.59	
500	44.58	
	44.57	
	44.56	
202	44.55	
	44.54	
421	44.53	
846	44.52	



Market Making

A market maker shows a *bid* and an *offer* i.e. a price at which (s)he wants to buy and a price at which he wants to sell the option

Example:

When a market maker believes an option's value is 44.60 eur, (s)he might want to buy this call option at 44.58 eur and sell it at 44.62 eur

A market maker does *not* want to take a *directional* position, unlike directional market participants, but tries to *buy and sell* the same product (many times) while capturing the money between the "bid-ask spread"

Hedge



Immediately after a market maker has done a buy or sell trade in an option, the price of the option might move because of price changes in the underlying asset

In order to avoid this price risk after an option trade, a market maker "neutralizes" this price risk by doing a *hedge*

This hedge is a trade in another instrument than the option itself, mostly the underlying asset (e.g. the stock), which generates the 'opposite profit / loss' as the option trade does when prices change

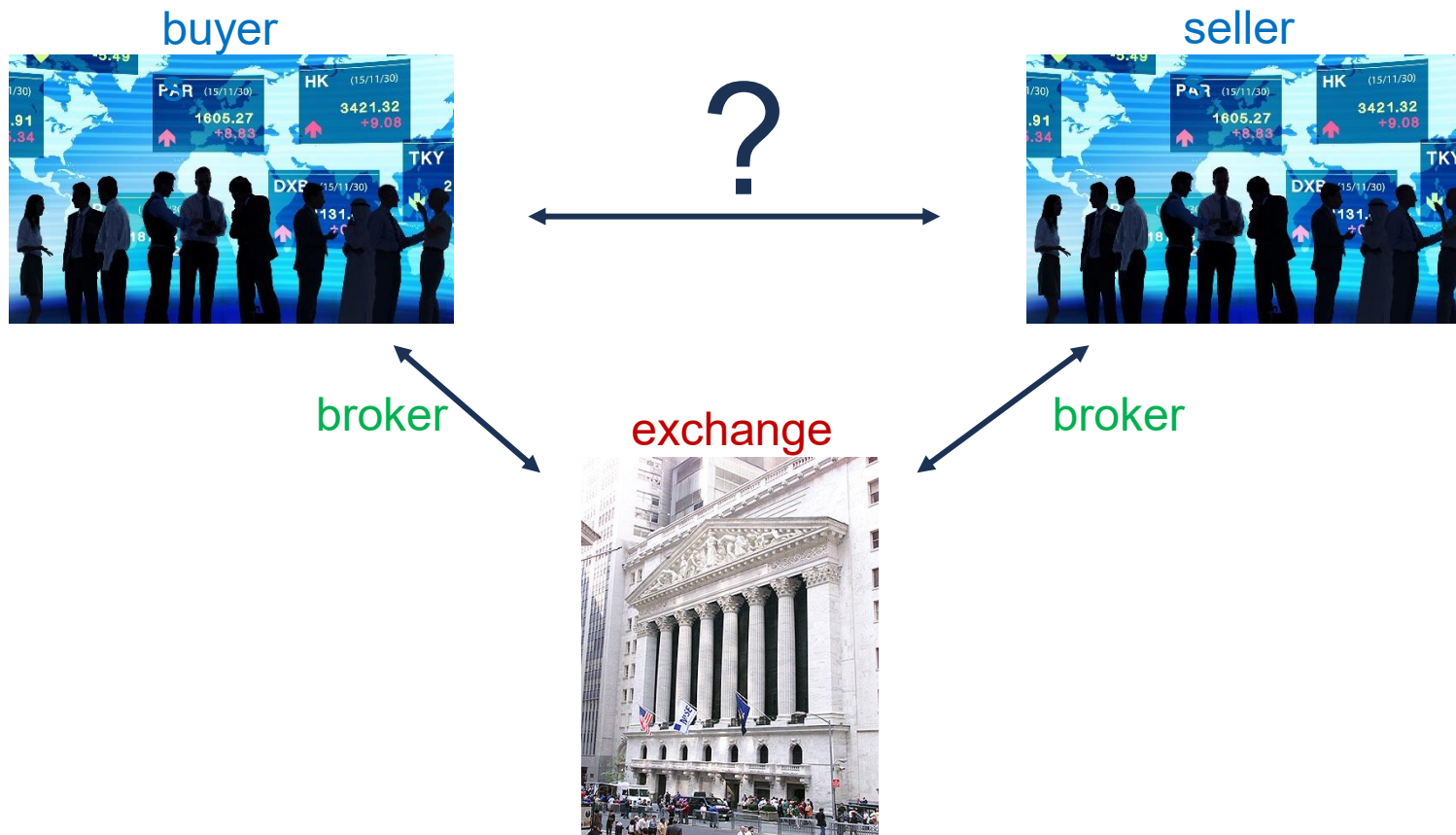
After the hedge is done we don't have directional exposure to the underlying asset anymore, i.e. non-directional



Trading and Execution

- Exchanges
- Order books
- Order book dynamics
- Order types
- Execution

Exchanges





Order Books

Price-time priority: orders with a higher buy price (or lower sell price) have higher priority than other orders. If more participants provide these best prices in the order book the time of arrival determines the priority.

bid volume	price	ask volume
	44.67	1230
	44.66	500
	44.65	611
	44.64	250
	44.63	310
	44.62	146
	44.61	
12	44.60	
156	44.59	
201	44.58	
182	44.57	
450	44.56	
520	44.55	
513	44.54	
790	44.53	
1400	44.52	

best bid →

best ask / best offer

Bid-ask spread
(44.62 - 44.60 = 0.02)

↑ bids ↑ price ladder ↑ offers



Order Book Dynamics

Dime / improve: you add an order which becomes the best bid (best offer) in the order book (e.g. add a buy order at 44.61)

Hit the bid: you sell at the existing highest bid price in the order book

Join: add your buy order to the existing volume (i.e. the existing volume of 450 lots will increase with your added order size)

bid volume	price	ask volume
	44.67	1230
	44.66	500
	44.65	611
	44.64	250
	44.63	310
	44.62	146
	44.61	
12	44.60	
156	44.59	
201	44.58	
182	44.57	
450	44.56	
520	44.55	
513	44.54	
790	44.53	
1400	44.52	

Pull / cancel: you take your existing order out of the order book

Lift the offer: you buy at the existing lowest ask price in the order book

Spoofing: sending (and subsequently pulling) orders to the order book without having the intention to execute these (fake) orders. This is NOT allowed!

Order Types



Exchanges and brokers often facilitate many different order types with different characteristics and behavior. Below we discuss a useful selection:

- **Market Order:** No price condition; will be executed immediately at best available prices
- **Limit Order:** Price condition; will be executed at best available prices equal to or better than its price limits
- **Fill or Kill:** Price condition; 'entire and immediate execution' or cancellation
- **Immediate or Cancel:** Price condition; will be executed immediately to the extent possible, the remaining unexecuted volume will be cancelled
- **Quote:** one message that contains both buy and sell order (usually used by market makers)
- **Stop (Market) Order:** Upon reaching the 'stop price' this order will become active as a Market Order
- **Stop (Limit) Order:** Upon reaching the 'stop price' this order will become active as a Limit Order

Order Types



bid #	price	ask #
	155	54
	154	60
	153	53
	152	35
	151	
	150	
1	149	
13	148	
24	147	
47	146	
64	145	
56	144	

1) Market order

bid #	price	ask #
	155	54
	154	60
	153	53
	152	37
2	151	
	150	
1	149	
13	148	
24	147	
47	146	
64	145	
56	144	

2) Limit order @ 151

Start situation

bid #	price	ask #
	155	54
	154	60
	153	53
	152	37
	151	15
	150	3
1	149	
13	148	
24	147	
47	146	
64	145	
56	144	

Sending a buy order for 20 lots in 4 different ways:

- 1) Market order
- 2) Limit order @ 151
- 3) Fill-or-kill @ 151
- 4) Immediate-or-cancel @ 151

bid #	price	ask #
	155	54
	154	60
	153	53
	152	37
	151	15
	150	3
1	149	
13	148	
24	147	
47	146	
64	145	
56	144	

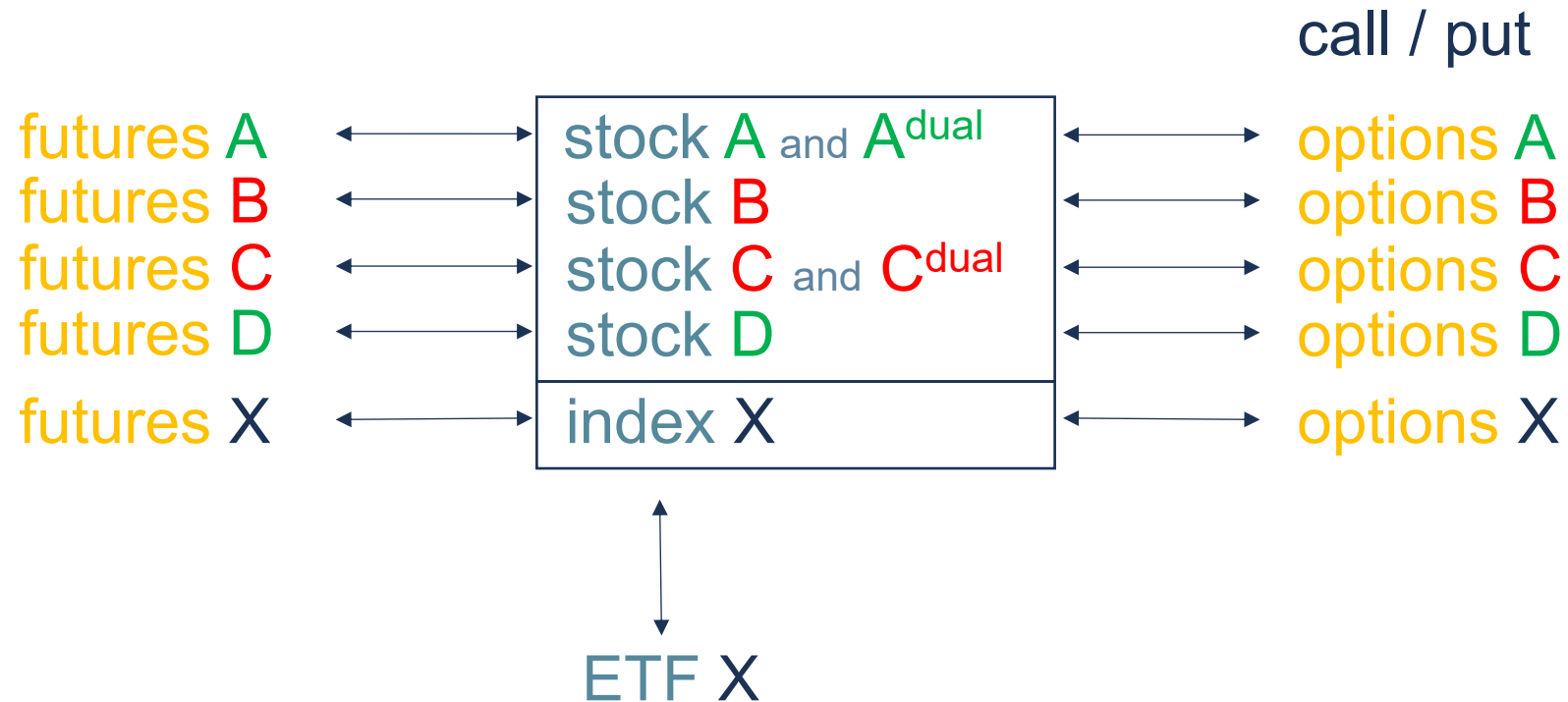
3) Fill-or-kill @ 151

bid #	price	ask #
	155	54
	154	60
	153	53
	152	37
	151	
	150	
1	149	
13	148	
24	147	
47	146	
64	145	
56	144	

4) IOC order @ 151



Dual / multiple stock listing





Execution

Stock ABC on exchange 1		
bid volume	price	ask volume
	44.64	1000
	44.63	1000
	44.62	1000
	44.61	1000
1000	44.60	
1000	44.59	
1000	44.58	
1000	44.57	

Stock ABC on exchange 2		
bid volume	price	ask volume
	44.64	164
	44.63	140
100	44.62	
	44.61	
214	44.60	
181	44.59	
200	44.58	
320	44.57	

A simple example of a trading strategy is that of a 'dual listing', i.e. one specific stock (of a company) is trading on two different exchanges (i.e. in two different order books).

Since it concerns the same stock (company) the value of the stocks should be the same. A difference in their price triggers a trading opportunity.



Execution – Example (1)

Price differences

Assuming that you can buy and sell as many stocks as you want, without transaction costs, in both order books, would you like to do a trade?

If so, which volumes do you buy and sell and in which order books?

Stock ABC on exchange 1		
bid volume	price	ask volume
	44.64	1000
	44.63	1000
	44.62	1000
	44.61	1000
1000	44.60	
1000	44.59	
1000	44.58	
1000	44.57	

2nd: Buy 100 stocks @ 44.61

Stock ABC on exchange 2		
bid volume	price	ask volume
	44.64	164
	44.63	140
100	44.62	
	44.61	
214	44.60	
181	44.59	
200	44.58	
320	44.57	

1st: Sell 100 stocks @ 44.62



Execution – Example (2)

Transaction fees

Same question, but now on both exchanges a transaction fee of 0.006 euro per traded stock is being charged. Do you still want to do a trade?

Stock ABC on exchange 1		
bid volume	price	ask volume
	44.64	1000
	44.63	1000
	44.62	1000
	44.61	1000
1000	44.60	
1000	44.59	
1000	44.58	
1000	44.57	

Stock ABC on exchange 2		
bid volume	price	ask volume
	44.64	164
	44.63	140
100	44.62	
	44.61	
214	44.60	
181	44.59	
200	44.58	
320	44.57	

No trade, because the costs are higher than the profit



Execution – Example (3)

Transaction fees

Again same question, but now on both exchanges a transaction fee of 0.70 euro per trade is being charged. Do you still want to do a trade?

Stock ABC on exchange 1		
bid volume	price	ask volume
	44.64	1000
	44.63	1000
	44.62	1000
	44.61	1000
1000	44.60	
1000	44.59	
1000	44.58	
1000	44.57	

Stock ABC on exchange 2		
bid volume	price	ask volume
	44.64	164
	44.63	140
100	44.62	
	44.61	
214	44.60	
181	44.59	
200	44.58	
320	44.57	

No trade, because the costs are higher than the profit

Trading terminology

- Order, Trade and Position
- Long vs Short
- Theoretical price
- Edge & Credit
- Pnl
- Hedge
- Bid, Ask and Mid price
- Last price
- Open and Close price
- Active vs Passive trading
- Aggressor vs Market Maker
- Order book & Price book
- Products, instruments and lots

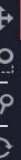


Leaderboard

Rank	Teams	PnL	Hourly PnL	Hourly Sharpe	Position	Delta
1	team-005	452,485.30	24.40	-0.01	48	48.00
2	team-006	40,948.70	120.70	0.03	71	71.00
3	team-008	2,144.30	156.20	0.06	23	23.00
4	team-011	2,030.00	33.00	0.05	0	0.00
5	team-007	1,015.30	-136.00	-0.08	-80	-80.00
6	team-017	343.10	131.20	0.08	133	133.00
7	team-010	255.50	277.80	0.10	122	122.00
8	team-009	205.10	279.70	0.09	105	105.00
9	team-002	74.60	-36.70	-0.04	25	25.00
10	team-016	1.10	0.60	-0.00	1	1.00
11	team-029	-17.50	-7.30	-0.09	-3	-3.00
12	team-003	-55.00	-0.90	0.02	-7	-7.00
13	team-013	-99.40	-35.20	-0.08	-11	-11.00
14	team-004	-144.00	106.70	0.05	0	0.00
15	team-001	-240.40	87.50	0.00	34	34.00

Instrument Overview

Instrument	# Bid	Best Bid	Best Ask	# Ask	Last Traded Price
CSCO	1,325	44.90	45.00	625	45.10
ING	500	115.60	115.70	825	115.50
NVDA	200	69.00	69.30	75	69.30
NVDA_DUAL	50	69.50	70.00	150	69.40
OB5X_ETF	750	39.10			39.20
OB5X_202309_F	50	149.00	149.20	150	149.00
OB5X_202312_F	50	150.20	150.30	150	150.20
OB5X_202403_F	50	151.30	151.50	150	151.30
PFE	200	9.30	9.40	700	9.40
SAN	1,325	44.60	44.70	200	44.80
SAN_DUAL	50	44.20	44.70	99	44.60



Trade Ticker

Time	Instrument	Price	Volume	Aggressor	Buyer	Seller
10:30:15	OB5X_202403_F	151.30	50	seller	PowerfulPineapples	RadiantRisottos
10:30:15	NVDA_DUAL	69.40	100	buyer	EarnestEspressos	DiligentDonuts
10:30:15	OB5X_202309_F	149.00	50	seller	PowerfulPineapples	RadiantRisottos
10:30:15	NVDA	69.30	425	seller	YieldingYams	AverageAlmonds
10:30:14	OB5X_202312_F	150.20	50	seller	PowerfulPineapples	RadiantRisottos
10:30:14	OB5X_ETF	39.20	203	buyer	KindlyKales	MarvelousMilkshakes
10:30:14	SAN_DUAL	44.70	1	buyer	GroovyGrapefruits	HomelyHerrings
10:30:14	SAN_DUAL	44.60	55	buyer	GroovyGrapefruits	Infinitelcecreams
10:30:14	SAN	44.80	200	seller	WavyWalnuts	YieldingYams
10:30:14	PFE	9.40	125	seller	YieldingYams	AverageAlmonds
10:30:14	NVDA	69.30	200	seller	YieldingYams	WavyWalnuts
10:30:13	SAN_DUAL	44.60	50	buyer	JazzyJellies	Infinitelcecreams
10:30:13	PFE	9.40	500	buyer	YieldingYams	AverageAlmonds
10:30:13	SAN_DUAL	44.20	50	seller	HomelyHerrings	GroovyGrapefruits
10:30:13	SAN_DUAL	44.60	45	buyer	GroovyGrapefruits	Infinitelcecreams



Optibook introduction

- URL
- Visualizer
- Sign up / login procedure
- Documentation
- Code templates
- Jupiter notebook
- Code_snippets.ipynb
- Dual listing trading strategy
- Technical issues: optibook@optiver.com



Thank you!



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